

Household Fuel Price Risk and the Uneven Protective Value of Domestic Net Zero: Evidence from Westminster, London

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Abstract

Fuel poverty identifies current affordability hardship, but it does not by itself measure future household exposure to fossil-fuel price shocks. This paper develops the concept of household fuel price risk and applies it to Westminster, London, using property-level or property-archetype modelling. The empirical strategy estimates required energy demand, baseline energy burden, price-shock sensitivity, and the distributional effect of domestic low-carbon interventions. The central argument is that domestic net zero is not automatically a household risk-protection strategy. Retrofit, clean heat, solar PV, battery storage, smart tariffs, and flexibility reduce unequal exposure only when housing fabric, heating systems, tariff design, market reform, tenure, and local governance align.

Keywords: fuel poverty; energy burden; price risk; domestic retrofit; net zero; Westminster; energy justice

1 Introduction

This paper reframes household energy inequality from a static problem of current fuel poverty into a dynamic problem of uneven exposure to future fossil-fuel price volatility. Westminster is an analytically useful case because high average wealth can obscure fuel-price vulnerability among households facing high housing costs, inefficient dwellings, private rental constraints, leasehold governance, heritage restrictions, and limited access to low-carbon assets.

The central research question is: to what extent, and under what housing, market, and governance conditions, do domestic net zero interventions reduce unequal household exposure to fossil-fuel price volatility in Westminster?

The paper addresses three linked questions:

1. Which household-dwelling types face the largest required energy burden increase under fuel price shocks?
2. How do income, housing fabric, heating system, tenure, and vulnerability jointly produce fuel price risk?
3. How much risk reduction do retrofit, clean heat, PV, battery storage, smart tariffs, and integrated packages deliver, and for whom?

The contribution is to evaluate domestic net zero interventions as conditional household de-risking tools rather than as automatic bill-reduction devices.

2 Conceptual Framework

The paper separates three concepts that are often conflated. Fuel poverty identifies present affordability hardship under current prices and required energy needs. Energy burden measures

required or actual energy cost relative to residual household income. Household fuel price risk captures future exposure, sensitivity, and limited adaptive capacity under price-shock scenarios.

$$\text{Household fuel price risk} = \text{exposure} + \text{sensitivity} - \text{adaptive capacity}. \quad (1)$$

This distinction matters because decarbonisation is not the same as de-risking. Electrification can reduce emissions before it reduces household price risk where electricity prices remain linked to gas through wholesale market design. The protective value of heat pumps, PV, storage, smart tariffs, and flexibility therefore depends on the electricity-gas price ratio, tariff access, capital access, dwelling form, tenure, and governance.

3 Case and Data

The empirical case is Westminster, London. The preferred unit of analysis is the property or household-dwelling pair. Where household microdata are unavailable, the study uses EPC property records combined with LSOA/MSOA income and deprivation proxies to assign synthetic households or archetypes.

The current enriched property master contains 124,980 properties across 126 LSOAs and 26 MSOAs. It combines EPC attributes, spatial identifiers, MSOA after-housing-cost income, LSOA income deprivation, official LSOA fuel poverty rates, and LSOA income-distribution outputs from the latest verified pipeline.

The analysis uses required energy demand rather than only observed expenditure, because actual consumption may understate unmet need through underheating or self-rationing.

4 Methods

The empirical strategy is a property-level income–cost and scenario model. It links EPC-derived dwelling characteristics, synthetic LSOA income distributions, required energy costs, price-shock scenarios, domestic net zero intervention packages, and spatial-distributional outputs. The unit of analysis is property p in LSOA l and MSOA m . Where observed household incomes are unavailable, the model estimates the probability that a household drawn from the local residual-income distribution would face a given required-energy burden. This design is deliberately probabilistic: it does not claim to observe the income of the current occupant of each property.

4.1 Income–Cost Engine

The implemented income layer estimates one constrained lognormal after-housing-cost income distribution for each LSOA:

$$Y_l \sim \text{Lognormal}(\mu_l, \sigma_l^2), \quad (2)$$

where Y_l denotes annual residual household income. The distribution is parameterised by its mean $\bar{Y}_l$:

$$\mu_l = \ln(\bar{Y}_l) - \frac{1}{2}\sigma_l^2. \quad (3)$$

For each MSOA, the LSOA parameters are jointly estimated so that the property-weighted mean remains anchored to the MSOA after-housing-cost income, while the lower tail is differentiated by the LSOA income-deprivation signal. For a given regularisation parameter λ , the objective is:

$$\begin{aligned} \mathcal{L}_m(\lambda) = & \left[\ln \left(\frac{\sum_{l \in m} N_l \bar{Y}_l}{\sum_{l \in m} N_l} \right) - \ln(\bar{Y}_m) \right]^2 \\ & + \lambda \sum_{l \in m} w_l [F_l(z) - d_l]^2, \end{aligned} \quad (4)$$

where N_l is the number of EPC property records in LSOA l , $w_l = N_l / \sum_{l \in m} N_l$, $\bar{Y}_m$ is MSOA residual income, $z = 19,009$ is the income threshold used in the current module, $F_l(\cdot)$ is the LSOA lognormal cumulative distribution function, and d_l is the LSOA income-deprivation score. The current implementation constrains $\sigma_l \in [0.10, 1.40]$ and searches a log-spaced grid of 100 λ values from 10^{-4} to 10^{-1} . Official LSOA fuel-poverty rates are used to select λ by minimising property-weighted RMSE, with weighted MAE and then the smaller λ as tie-breakers; they are not entered directly as the optimisation target.

Required energy cost is derived from SAP and floor area rather than from observed consumption, because actual expenditure may be suppressed by underheating or self-rationing. For property p , the SAP energy cost factor is:

$$\text{ECF}_p = \begin{cases} (100 - \text{SAP}_p)/16.21, & \text{SAP}_p \geq 43.265, \\ 10^{(108.8 - \text{SAP}_p)/120.5}, & \text{SAP}_p < 43.265. \end{cases} \quad (5)$$

The baseline required annual energy cost is then:

$$C_{p,0} = \text{ECF}_p \frac{A_p + 45}{0.2}, \quad (6)$$

where A_p is total floor area. Baseline required energy burden for a household with income Y_l is:

$$B_{p,0}(Y_l) = \frac{C_{p,0}}{Y_l}. \quad (7)$$

The current module therefore estimates burden-threshold probabilities:

$$P_{p,0}(\tau) = \Pr(B_{p,0}(Y_l) > \tau) = F_l\left(\frac{C_{p,0}}{\tau}\right). \quad (8)$$

The implemented baseline output uses $\tau = 0.10$. For modelled LSOA fuel-poverty comparison, properties with SAP grades A, B, or C are assigned a property poverty score of zero; other properties use $P_{p,0}(0.10)$. This score is a baseline affordability input, not a complete measure of household fuel price risk.

4.2 Price-Shock Risk Metrics

The price-shock layer converts the baseline income-cost engine into forward-looking risk metrics. Let $s \in S$ denote a fuel-price scenario, including gas-price, electricity-price, combined-fuel, standing-charge, and policy-mitigation cases. Required cost under scenario s is written as:

$$C_{p,s} = \pi_s^g Q_p^g + \pi_s^e Q_p^e + SC_s^g I_p^g + SC_s^e I_p^e - M_{p,s}, \quad (9)$$

where Q_p^g and Q_p^e are required gas and electricity quantities or archetype-imputed equivalents, π_s^g and π_s^e are scenario fuel prices, SC_s^g and SC_s^e are standing charges, I_p^g and I_p^e indicate fuel connections, and $M_{p,s}$ captures bill support or other mitigation. The scenario burden is:

$$B_{p,s}(Y_l) = \frac{C_{p,s}}{Y_l}, \quad (10)$$

and the burden uplift is:

$$\Delta B_{p,s}(Y_l) = B_{p,s}(Y_l) - B_{p,0}(Y_l). \quad (11)$$

The main risk indicators are defined as distributional rather than average outcomes. The scenario-specific threshold-crossing probability is:

$$P_{p,s}(\tau) = F_l\left(\frac{C_{p,s}}{\tau}\right). \quad (12)$$

For scenario weights ω_s , the expected threshold-crossing risk is:

$$R_p(\tau) = \sum_{s \in S} \omega_s P_{p,s}(\tau). \quad (13)$$

An affordability gap-at-risk can be expressed as the expected shortfall above an acceptable burden threshold:

$$G_{p,s}(\tau) = E_{Y_l} [\max\{C_{p,s} - \tau Y_l, 0\}]. \quad (14)$$

Household Fuel Price-at-Risk is defined as a high quantile of the scenario burden distribution:

$$\text{HFPaR}_{p,\alpha} = \inf \left\{ b : \sum_{s \in S} \omega_s \mathbf{1}(B_{p,s} \leq b) \geq \alpha \right\}, \quad (15)$$

where α may be set to a policy-relevant upper-tail level such as 0.90 or 0.95. These metrics keep fuel poverty, energy burden, and fuel price risk analytically separate: the first concerns current hardship, the second is a cost-to-income ratio, and the third measures future exposure and sensitivity under price shocks.

4.3 Intervention Counterfactuals

Domestic net zero interventions are represented as counterfactual changes to required demand, fuel structure, tariff exposure, and mitigation access. Let k denote an intervention package, such as fabric retrofit, clean heat, PV, battery storage, smart tariffs, or an integrated package. The post-intervention scenario cost is:

$$C_{p,s,k} = a_{p,k} \tilde{C}_{p,s,k} + (1 - a_{p,k}) C_{p,s}, \quad (16)$$

where $a_{p,k} \in [0, 1]$ represents technical, tenure, leasehold, heritage, roof, finance, or tariff feasibility. This term is central to the paper's interpretation: an intervention produces household risk protection only where access and market conditions allow the technical change to translate into lower exposure.

The intervention-specific threshold risk and Fuel Price-at-Risk are:

$$P_{p,s,k}(\tau) = F_l \left(\frac{C_{p,s,k}}{\tau} \right), \quad \text{HFPaR}_{p,\alpha,k} = Q_\alpha (\{B_{p,s,k}\}_{s \in S}). \quad (17)$$

The de-risking dividend is the reduction in upper-tail burden risk:

$$D_{p,\alpha,k} = \text{HFPaR}_{p,\alpha,0} - \text{HFPaR}_{p,\alpha,k}. \quad (18)$$

Results will be stratified by income distribution, EPC rating, dwelling form, heating system, tenure, leasehold and heritage constraints, and LSOA/MSOA location. The empirical question is therefore not whether a low-carbon measure reduces emissions in engineering terms, but who receives measurable price-risk protection once required demand, tariffs, fuel-market pass-through, and local governance constraints are jointly considered.

5 Expected Outputs

The empirical outputs will include LSOA/MSOA maps, property-archetype summaries, distributional plots, and intervention comparisons. A central output is the de-risking dividend:

$$\text{De-risking dividend} = \text{Fuel Price-at-Risk}_{\text{before}} - \text{Fuel Price-at-Risk}_{\text{after}}. \quad (19)$$

The paper will also identify de-risking gaps: places and household-dwelling types where fuel price protection is most needed but low-carbon intervention is least accessible.

6 Discussion

The expected policy implication is that domestic net zero should be treated as a distributional household risk-protection agenda. Protection depends on the alignment of fabric retrofit, heating technology, distributed generation, flexibility, tariff design, electricity market reform, capital access, and tenure-sensitive governance.

In Westminster, this means paying particular attention to private renters, leasehold flats, HMOs, heritage-constrained dwellings, district-heating governance, social housing delivery pathways, and households whose vulnerability is masked by high neighbourhood average wealth.

7 Conclusion

Fuel poverty identifies current hardship; household fuel price risk identifies future exposure to price shocks. Net zero determines who receives protection only when domestic interventions are technically effective, financially accessible, and institutionally deliverable for the households most exposed.

The paper therefore asks not whether net zero reduces emissions, but whether, how, and for whom it credibly reduces household exposure to fossil-fuel price volatility.